

Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.19)

Metric

$$ds^2 = (-\lambda z^2 - 1) dt \otimes dt + (-\lambda y^2 + 1) dx \otimes dx + \left(-\frac{1}{\lambda y^2 - 1}\right) dy \otimes dy + \left(\frac{1}{\lambda z^2 + 1}\right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}\lambda z^2 + \frac{1}{2}}\right) dt + \left(\frac{1}{\sqrt{2\lambda z^2 + 2}}\right) dz$$

$$n = \left(\sqrt{\frac{1}{2}\lambda z^2 + \frac{1}{2}}\right) dt + \left(-\frac{1}{\sqrt{2\lambda z^2 + 2}}\right) dz$$

$$m = \left(\sqrt{-\frac{1}{2}\lambda y^2 + \frac{1}{2}}\right) dx + \left(\frac{i}{\sqrt{-2\lambda y^2 + 2}}\right) dy$$

$$\bar{m} = \left(\sqrt{-\frac{1}{2}\lambda y^2 + \frac{1}{2}}\right) dx + \left(-\frac{i}{\sqrt{-2\lambda y^2 + 2}}\right) dy$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\sqrt{\lambda z^2 + 1}} \right) dt + \left(\frac{1}{2} \sqrt{2}\sqrt{\lambda z^2 + 1} \right) dz$$

$$n = \left(-\frac{\sqrt{2}}{2\sqrt{\lambda z^2 + 1}} \right) dt + \left(-\frac{1}{2} \sqrt{2}\sqrt{\lambda z^2 + 1} \right) dz$$

$$m = \left(-\frac{\sqrt{2}}{2\sqrt{\lambda y^2 - 1}} \right) dx + \left(-\frac{1}{2} i \sqrt{2}\sqrt{\lambda y^2 - 1} \right) dy$$

$$\bar{m} = \left(-\frac{\sqrt{2}}{2\sqrt{\lambda y^2 - 1}} \right) dx + \left(\frac{1}{2} i \sqrt{2}\sqrt{\lambda y^2 - 1} \right) dy$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\sqrt{\lambda z^2 + 1}} \partial_t + \frac{1}{2} \sqrt{2}\sqrt{\lambda z^2 + 1} \partial_z$$

$$\Delta = -\frac{\sqrt{2}}{2\sqrt{\lambda z^2 + 1}} \partial_t - \frac{1}{2} \sqrt{2}\sqrt{\lambda z^2 + 1} \partial_z$$

$$\delta = -\frac{\sqrt{2}}{2\sqrt{\lambda y^2 - 1}} \partial_x - \frac{1}{2} i \sqrt{2}\sqrt{\lambda y^2 - 1} \partial_y$$

$$\bar{\delta} = -\frac{\sqrt{2}}{2\sqrt{\lambda y^2 - 1}} \partial_x + \frac{1}{2} i \sqrt{2}\sqrt{\lambda y^2 - 1} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}\lambda z}{4\sqrt{\lambda z^2 + 1}}$$

$$\gamma = \frac{\sqrt{2}\lambda z}{4\sqrt{\lambda z^2 + 1}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

$$NP6 = -\frac{1}{2}\lambda$$

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"